

2. Cardinality, countability, and non-negative sums

REAL VARIABLES I
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2.1 Cardinality and countability

Definition 2.1.1 (Finite, infinite, and countable sets). A set A is *finite* if there are $n \in \mathbb{N}$ and a bijection $\iota : \{1, \dots, n\} \rightarrow A$. It is *infinite* if it is not finite.

A set A is *countably infinite* if there is a bijection $\iota : \mathbb{N} \rightarrow A$. It is *countable*, or *at most countable*, if it is finite or countably infinite.

2.1.1 Digression on cardinality

Definition 2.1.2 (Comparison of cardinalities). Two sets A and B have the same cardinality, written $A \sim B$ or $|A| = |B|$, if there is a bijection $A \rightarrow B$.

Write $|A| \leq |B|$ if there is an injection $A \rightarrow B$. Write $|A| \geq |B|$ if there is a surjection $A \rightarrow B$.

Both these relations are reflexive, and transitive. One can show that the existence of an injection $A \rightarrow B$ is equivalent to the existence of a surjection $B \rightarrow A$ (this, I believe, uses the axiom of choice). The next theorem shows that $|A| \leq |B|$ is indeed an order relation.

Theorem (Cantor–Bernstein theorem). Let A and B be sets. If there are injections $f : A \rightarrow B$ and $g : B \rightarrow A$, then there is a bijection $A \rightarrow B$.

The proof is outside the scope of these notes.

2.1.2 Countable sets

We are mainly concerned with at-most-countable sets. An important fact to know is that *countable infinite sets* are the smallest infinite cardinality:

In practice, we record the following useful facts.

Proposition ($|\mathbb{N}|$ is the smallest infinite cardinality). If a set A is infinite, then there exists an injection $\iota : \mathbb{N} \rightarrow A$, i.e. $|\mathbb{N}| \leq |A|$.

Proposition (Characterizations of countability). For a set A , the following are equivalent:

1. A is countable;
2. either $A = \emptyset$, or there exists a countable B and a surjection $B \rightarrow A$.
3. there exists a countable B and an injection $A \rightarrow B$;

Corollary 2.1.3 (Countable unions and finite products of countable sets).

Let $(A_n)_{n \in \mathbb{N}}$ be a sequence of countable sets. Then $\bigcup_{n \in \mathbb{N}} A_n$ is countable.

Let $m \in \mathbb{N}$, and let $A_1, \dots, A_m$ be countable sets. Then $A_1 \times \dots \times A_m$ is countable.

Proof. (proves 2.1.3)

First, let us show that $\mathbb{N} \times \mathbb{N}$ is countable. Indeed, the map $\pi: \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$, given by $\pi(r, s) = 2^r(2s + 1)$, is a bijection with $\mathbb{N} \setminus \{0\}$ (put the first number into the power of 2 and then multiply by an odd number).

To show that countable unions of countable sets are countable we construct a surjective map from $\mathbb{N} \times \mathbb{N}$ to $\bigcup_{n \in \mathbb{N}} A_n$: for each n , choose a surjection $\iota_n: \mathbb{N} \rightarrow A_n$ and define $(n, m) \mapsto \iota_n(m)$. It is a surjection and $\mathbb{N} \times \mathbb{N}$ is countable.

To show countability of $A \times B$ when A and B are countable, choose surjections

$$\alpha: \mathbb{N} \rightarrow A, \quad \beta: \mathbb{N} \rightarrow B.$$

and define the map on $\mathbb{N} \times \mathbb{N}$ given by $(n, m) \rightarrow (\alpha(n), \beta(m))$. It is a surjection and $\mathbb{N} \times \mathbb{N}$ is countable. $\square$

Corollary 2.1.4 (Countability of the rational numbers). The set $\mathbb{Q}$ is countable.

Indeed, the set $\mathbb{Z}$ is countably infinite, as witnessed by the enumeration

$$0, 1, -1, 2, -2, \dots$$

Therefore $\mathbb{Z} \setminus \{0\}$ and $\mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$ are countable, The map $(m, n) \mapsto \frac{m}{n}$ is a surjection from this last product onto $\mathbb{Q}$. A surjective image of a countable set is countable.

Theorem 2.1.5 (Uncountability of binary sequences). The set $\{0, 1\}^{\mathbb{N}} = \{(x_n)_{n \in \mathbb{N}} : x_n \in \{0, 1\}\}$ is uncountable.

Proof. (proves 2.1.5)

Suppose that $\{0, 1\}^{\mathbb{N}}$ is countable. Since it is nonempty, there exists a surjection $F: \mathbb{N} \rightarrow \{0, 1\}^{\mathbb{N}}$. Write $F(k) = (x_n^{(k)})_{n \in \mathbb{N}}$. Define $y = (y_n)_{n \in \mathbb{N}}$ by $y_n = 1 - x_n^{(n)}$. Then $y \in \{0, 1\}^{\mathbb{N}}$, but $y \neq F(k)$ for every k , since their k -th coordinates differ. This contradicts surjectivity of F . $\square$

Corollary 2.1.6 (Uncountability of the real numbers). The set $\mathbb{R}$ is uncountable.

Proof. (proves 2.1.6)

Define the base 3 expansion operation $\Phi: \{0, 1\}^{\mathbb{N}} \rightarrow [0, 1]$ by

$$\Phi(x) = \sum_{n=1}^{\infty} \frac{x_n}{3^n}.$$

This is the ternary expansion obtained by using only the digits 0 and 1. This map is injective. Indeed, suppose $x \neq y$ and let n_0 be the first digit where they differ. By symmetry suppose $x_{n_0} = 1$ and $y_{n_0} = 0$. We have

$$\begin{aligned} \Phi(x) - \Phi(y) &= \frac{x_{n_0} - y_{n_0}}{3^{n_0}} + \sum_{k=n_0+1}^{\infty} \frac{x_k - y_k}{3^k} \geq \frac{1}{3^{n_0}} - \sum_{k=n_0+1}^{\infty} \frac{|x_k - y_k|}{3^k} \\ &\geq \frac{1}{3^{n_0}} - \sum_{k=n_0+1}^{\infty} \frac{2}{3^k} \geq \frac{1}{3^{n_0}} - \frac{2}{3^{n_0}} \frac{1}{1 - 1/3} = \frac{1}{3^{n_0}} - \frac{1}{3^{n_0+1}} > 0. \end{aligned}$$

The uncountability of $\{0, 1\}^{\mathbb{N}}$ from Theorem 2.1.5 implies that $[0, 1]$, and therefore $\mathbb{R}$, is uncountable. $\square$

2.2 Extended reals and nonnegative sums

Definition 2.2.1 (Extended real numbers). The extended real numbers are $\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}$, ordered by $-\infty < x < +\infty$ for every $x \in \mathbb{R}$. The extended nonnegative real numbers are $\overline{\mathbb{R}}_+ = [0, +\infty]$.

Addition and multiplication on $\overline{\mathbb{R}}_+$ extend the usual operations by $a + (+\infty) = (+\infty) + a = +\infty$

$$a \cdot (+\infty) = (+\infty) \cdot a = +\infty \quad \text{for } a > 0$$

with the convention

$$0 \cdot (+\infty) = (+\infty) \cdot 0 = 0.$$

Other operations are not defined.

Every subset of $\overline{\mathbb{R}}$ has a supremum and an infimum in $\overline{\mathbb{R}}$, with

$$\sup \emptyset = -\infty, \quad \inf \emptyset = +\infty.$$

Warning (Undefined extended-real subtraction). The expression $+\infty + (-\infty)$ is undefined. An argument involving extended-real quantities may use subtraction only after the relevant terms have been shown to be finite or the expression has otherwise been given a separate definition.

2.2.1 Arbitrary nonnegative sums

Definition 2.2.2 (Arbitrary nonnegative sum). Let $\mathcal{J}$ be any index set, and let $(a_j)_{j \in \mathcal{J}}$ be a family in $\overline{\mathbb{R}}_+$. Define

$$\sum_{j \in \mathcal{J}} a_j = \sup \left\{ \sum_{j \in F} a_j : F \subseteq \mathcal{J} \text{ is finite} \right\}.$$

The sum over the empty set is 0. For $K \subseteq \mathcal{J}$, the notation $\sum_{j \in K} a_j$ denotes the sum of the *finite* family $(a_j)_{j \in K}$.

The definition immediately gives the following monotonicity properties:

- if $K \subseteq L \subseteq \mathcal{J}$ then $\sum_{j \in K} a_j \leq \sum_{j \in L} a_j$,
- if $0 \leq a_j \leq b_j$ for every j then $\sum_{j \in \mathcal{J}} a_j \leq \sum_{j \in \mathcal{J}} b_j$.

Furthermore, we will recurrently use the following approximation property that follows immediately from the definition of sup.

- If $S = \sum_{j \in \mathcal{J}} a_j < +\infty$, then for every $\varepsilon > 0$ there is a finite $F \subseteq \mathcal{J}$ such that $S - \varepsilon < \sum_{j \in F} a_j \leq S$.
- If $S = +\infty$, then for every $M > 0$ there is a finite $F \subseteq \mathcal{J}$ such that $\sum_{j \in F} a_j > M$.

The next theorem shows that any finite sum is supported on a countable set.

Theorem 2.2.3 (Countable support of a summable family). Let $(a_j)_{j \in \mathcal{J}}$ be a family in $\overline{\mathbb{R}}_+$. If $\sum_{j \in \mathcal{J}} a_j < +\infty$, then $\{j \in \mathcal{J} : a_j \neq 0\}$ is countable.

Proof. (proves 2.2.3)

For $n \in \mathbb{N}$, set $E_n = \{j \in \mathcal{J} : a_j \geq 2^{-n}\}$. Each E_n is finite because otherwise, for every $m \in \mathbb{N}$ one could choose m distinct elements of E_n , obtaining $\sum_{j \in \mathcal{J}} a_j \geq m2^{-n}$. Since m is arbitrary, the sum would be $+\infty$.

Every j with $a_j > 0$ belongs to E_n for some n . Hence $\{j \in \mathcal{J} : a_j \neq 0\} = \bigcup_{n \in \mathbb{N}} E_n$, a countable union of countable sets, which is countable. $\square$

Next, we show a continuity property from below of these sums. We say a collection of subsets $(K_n \subset \mathcal{J})_{n \in \mathbb{N}}$ is non-decreasing if $K_1 \subseteq K_2 \subseteq \dots$. We write $K_n \uparrow \mathcal{J}$ if $(K_n \subset \mathcal{J})_{n \in \mathbb{N}}$ is non-decreasing and $\bigcup_{n=1}^{\infty} K_n = \mathcal{J}$. This is analogous to writing $x_n \uparrow z$ for a sequence of real numbers: it means $x_{n+1} \geq x_n$ for all n and $\lim_n x_n = z$. Note that for a monotone non-decreasing sequence, the limit always exists and $\lim_n x_n = \sup_n x_n$.

Proposition 2.2.4 (Continuity under increasing index sets). Let $(a_j)_{j \in \mathcal{J}}$ be a family in $\bar{\mathbb{R}}_+$ and suppose $K_n \uparrow \mathcal{J}$. Then

$$\sum_{j \in K_n} a_j \uparrow \sum_{j \in \mathcal{J}} a_j.$$

The sets K_n need not be finite.

Proof. (proves 2.2.4)

Set $S_n = \sum_{j \in K_n} a_j$, and $S = \sum_{j \in \mathcal{J}} a_j$.

By monotonicity we have $S_n \leq S_{n+1} \leq S$. Taking the sup of the right-hand side we have

$$\lim_n S_n = \sup_n S_n \leq S.$$

It remains to show the converse inequality: $S \leq \sup_n S_n$.

If $S < +\infty$, for every $\varepsilon > 0$ choose a finite F_ε such that $\sum_{j \in F_\varepsilon} a_j > S - \varepsilon$. Since F_ε is finite and $K_n \uparrow \mathcal{J}$, there exists n_0 such that $F_\varepsilon \subset K_n$ for all $n \geq n_0$. This shows that $S_n \geq \sum_{j \in F_\varepsilon} a_j > S - \varepsilon$ for all n large enough and therefore, since $\varepsilon > 0$ is arbitrary, $\sup_n S_n \geq S$, as desired.

If $S = +\infty$ for every M choose a finite F_M such that $\sum_{j \in F_M} a_j > M$. Since F_M is finite and $K_n \uparrow \mathcal{J}$, there exists n_0 such that $F_M \subset K_n$ for all $n \geq n_0$. This shows that $S_n \geq \sum_{j \in F_M} a_j > M$ for all n large enough and therefore, since M is arbitrary, $\sup_n S_n = +\infty$. $\square$

Taking $\mathcal{J} = \mathbb{N}$ and $K_n = \{1, \dots, n\}$ in Proposition 2.2.4 gives

$$\sum_{j \in \mathbb{N}} a_j = \lim_{n \rightarrow \infty} \sum_{j=1}^n a_j.$$

Thus Definition 2.2.2 agrees with the usual definition of a nonnegative series.

Proposition 2.2.5 (Tonelli for nonnegative sums). Let A and B be arbitrary sets, and let $(a_{i,j})_{(i,j) \in A \times B}$ be a family in $\bar{\mathbb{R}}_+$. Then

$$\sum_{i \in A} \left(\sum_{j \in B} a_{i,j} \right) = \sum_{(i,j) \in A \times B} a_{i,j} = \sum_{j \in B} \left(\sum_{i \in A} a_{i,j} \right).$$

Proof. (proves 2.2.5)

We only prove $\sum_{i \in A} \left(\sum_{j \in B} a_{i,j} \right) = \sum_{(i,j) \in A \times B} a_{i,j}$. The other claim is symmetric.

First we show that

$$\sum_{i \in A} \left(\sum_{j \in B} a_{i,j} \right) \geq \sum_{(i,j) \in A \times B} a_{i,j}.$$

Indeed, if $F \subseteq A \times B$ is finite, then $F \subseteq \pi_A(F) \times \pi_B(F)$, where $\pi_A(F)$ is the projection of $F \subset A \times B$ onto A (π_B is analogous). Hence

$$\sum_{(i,j) \in F} a_{i,j} \leq \sum_{i \in \pi_A(F) \times \pi_B(F)} \sum_{j \in \pi_B(F)} a_{i,j} \leq \sum_{i \in \pi_A(F)} \sum_{j \in \pi_B(F)} a_{i,j} \leq \sum_{i \in \pi_A(F)} \sum_{j \in B} a_{i,j} \leq \sum_{i \in A} \sum_{j \in B} a_{i,j}.$$

For the reverse inequality, set $T = \sum_{(i,j) \in A \times B} a_{i,j}$. If $T = \infty$, there is nothing to prove. Suppose $T < \infty$ and, toward a contradiction, $\sum_{i \in A} \sum_{j \in B} a_{i,j} > T$. Then there is a finite set $I \subseteq A$ such that $\sum_{i \in I} \sum_{j \in B} a_{i,j} > T$. Since I is finite, for each $i \in I$ one may choose a finite set $J_i \subseteq B$ so that $\sum_{i \in I} \sum_{j \in J_i} a_{i,j} > T$. Setting $J = \bigcup_{i \in I} J_i$, we obtain

$$T \geq \sum_{(i,j) \in I \times J} a_{i,j} \geq \sum_{i \in I} \sum_{j \in J_i} a_{i,j} > T,$$

a contradiction, as desired. $\square$

2.2.2 Convergence theorems for nonnegative sums

We leave the proofs of the following two facts to the reader.

Proposition 2.2.6 (Monotone convergence for nonnegative sums). Let $\mathcal{J}$ be an arbitrary index set. For $j \in \mathcal{J}$ and $k \in \mathbb{N}$, let $a_{j,k} \in \bar{\mathbb{R}}_+$. Suppose that, for every $j \in \mathcal{J}$,

$$a_{j,k} \uparrow a_j \quad \text{as } k \rightarrow \infty.$$

Prove that

$$\sum_{j \in \mathcal{J}} a_{j,k} \uparrow \sum_{j \in \mathcal{J}} a_j.$$

We leave the proof as an exercise.

Proposition 2.2.7 (Dominated convergence for nonnegative sums). Let $\mathcal{J}$ be an arbitrary index set and $(a_{j,k} \in \bar{\mathbb{R}}_+)_{(j,k) \in \mathcal{J} \times \mathbb{N}}$. Suppose that

- $\lim_k a_{j,k} = a_j$ for each $j \in \mathcal{J}$;
- There exists a dominating profile $(g_j \in \bar{\mathbb{R}}_+)_{j \in \mathcal{J}}$ such that $\sum_{j \in \mathcal{J}} g_j < +\infty$, and

$$a_{j,k} \leq g_j \quad \text{for every } j \in \mathcal{J} \text{ and } k \in \mathbb{N}.$$

Then

$$\lim_{k \rightarrow \infty} \sum_{j \in \mathcal{J}} a_{j,k} = \sum_{j \in \mathcal{J}} a_j.$$